

ALGORITHMS USED FOR DATA TRAINING AND MODELING

Complete Reference Guide for Sketch Generation, NLP, and Speech Recognition

1. SKETCH GENERATION (Variational Autoencoder Model)

Neural Network Algorithms

- Variational Autoencoder (VAE)** - Main generative model for sketch images
- Convolutional Neural Networks (CNN)** - Image feature extraction from sketches
- Encoder Network** - Compresses input images to latent space representation
- Decoder Network** - Reconstructs images from latent vectors
- Reparameterization Trick** - Enables backpropagation through stochastic nodes

Loss Functions

- Binary Cross Entropy (BCE) Loss** - Reconstruction loss for pixel-level comparison
- Kullback-Leibler Divergence (KLD)** - Regularization loss for latent space distribution

Activation Functions

- ReLU (Rectified Linear Unit)** - Hidden layer activation
- Sigmoid** - Output layer activation for pixel values in range [0,1]

Optimization Algorithms

- Adam Optimizer** - Adaptive moment estimation for gradient descent optimization
- Backpropagation** - Gradient computation through network layers

Data Processing Algorithms

- Random Sampling** - Generating random latent vectors for sketch generation
- Image Normalization** - Scaling pixel values to range [0,1]
- Train-Test Split (80-20)** - Dataset partitioning for validation

2. NATURAL LANGUAGE PROCESSING (BERT Model)

Transformer Algorithms

- BERT (Bidirectional Encoder Representations from Transformers)** - Main pre-trained model
- Multi-Head Self-Attention** - Captures relationships between words in text
- Positional Encoding** - Adds sequence position information to embeddings
- Layer Normalization** - Stabilizes training across layers
- Feed-Forward Neural Networks** - Within transformer blocks for processing

Tokenization Algorithms

- WordPiece Tokenization** - Subword tokenization algorithm for text processing
- Padding** - Extends sequences to fixed length of 512 tokens
- Truncation** - Cuts longer sequences to maximum length

Training Algorithms

- Fine-Tuning (Transfer Learning)** - Adapting pre-trained BERT to sentiment analysis
- Softmax Classification** - Final layer for binary sentiment prediction
- Cross-Entropy Loss** - Classification error computation

Optimization and Regularization

- AdamW Optimizer** - Weight decay version of Adam optimizer
- Weight Decay** - L2 regularization technique for preventing overfitting
- Warmup Steps** - Learning rate scheduling for stable training
- Gradient Clipping** - Prevents exploding gradients during training

Evaluation Metrics

- Accuracy Score** - Percentage of correct sentiment predictions
- F1 Score** - Harmonic mean of precision and recall
- Confusion Matrix** - True/False positive and negative analysis

3. SPEECH RECOGNITION (MLP Model)

Feature Extraction Algorithms

- MFCC (Mel-Frequency Cepstral Coefficients)** - Primary audio feature extraction
- Short-Time Fourier Transform (STFT)** - Underlying computation for MFCC
- Mel Filter Bank** - Frequency warping to mel scale for human perception
- Discrete Cosine Transform (DCT)** - Cepstral coefficient calculation
- Hamming Window** - Windowing function for audio frame processing

Audio Processing Algorithms

- Librosa Load** - Audio file loading and resampling to 16kHz
- Zero Padding** - Sequence length normalization to 16000 samples
- Truncation** - Cutting longer audio sequences to fixed length
- Mean Pooling** - Averaging MFCC features across time frames

Neural Network Algorithms

- Multi-Layer Perceptron (MLP)** - Deep feedforward network architecture
- Fully Connected Layers (Linear Layers)** - Dense neural connections
- Batch Normalization** - Normalizes layer inputs for stable training
- Dropout (30%)** - Random neuron deactivation for regularization

Classification Algorithms

- Label Encoding** - Converting digit labels 0-9 to numerical values
- Argmax Prediction** - Selecting highest probability class
- Cross-Entropy Loss** - Multi-class classification loss for 10 digits

4. SHARED ALGORITHMS ACROSS ALL TASKS

Data Management Algorithms

- NumPy Array Operations** - Efficient numerical computations for all models
- Pandas DataFrame** - Tabular data handling and manipulation
- Random Sampling** - Data shuffling and selection for training

Visualization Algorithms

- Matplotlib Plotting** - Loss curves and accuracy graphs generation
- Seaborn Heatmap** - Confusion matrix visualization with color coding
- Subplot Layout** - Multi-chart arrangement for comparison

Regularization Algorithms

- L2 Regularization** - Via weight decay in AdamW optimizer
- Dropout** - Random neuron dropout to prevent overfitting
- Batch Normalization** - Internal covariate shift reduction

5. ALGORITHM SUMMARY TABLE

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